Some Potentially Useful Formulas and Definitions

- odds of success $= \dfrac{\text{number of successes}}{\text{number of failures}}$
- odds ratio of success (Group A relative to Group B) $= \dfrac{\text{odds of success in Group A}}{\text{odds of success in Group B}}$
- $\text{SE}(\hat{p}_1 - \hat{p}_2) = \sqrt{\dfrac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \dfrac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
- Confidence Interval $= (\textbf{statistic}) \pm (\textbf{margin of error})$
  $= (\textbf{statistic}) \pm (\textbf{multiplier}) \cdot (\textbf{standard deviation/error})$

You may use the remainder of this page for scratch work; please keep in mind that NOTHING written here will be graded.